Aerospace Medicine and Human Performance

Cold Atmospheric Pressure Plasma Jets for Wound Management in Long-Duration Spaceflight A Review

--Manuscript Draft--

Manuscript Number:	AMHP6893R1
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Abstract:	Abstract Background: Long-duration spaceflight, such as missions planned under NASA's Artemis program and to Mars, presents critical medical risks including impaired wound healing and infections from multidrug-resistant pathogens. The limitations of mass, volume, and medication shelf life make conventional wound care strategies inadequate. Cold Atmospheric Pressure Plasma Jets (CAPPJ) present a multifunctional, non-antibiotic therapeutic platform with demonstrated terrestrial efficacy. Objective: This review synthesizes the mechanisms of Cold Atmospheric Pressure Plasma (CAPP) action and terrestrial clinical evidence to evaluate its potential as an integrated medical countermeasure for wound management in spaceflight. Methods: A literature search was conducted. Search terms included "cold atmospheric plasma," "wound healing," "antimicrobial," "biofilms," "immunomodulation," and "space medicine." The review focuses on the bioactive components of CAPPJ, molecular to tissue-level mechanisms, and outcomes from clinical trials, contextualized within the constraints of spaceflight. Results: CAPPJ technology simultaneously eradicates pathogens, including biofilms, without inducing resistance; modulates dysregulated inflammation via pathways such as NRF2 and NF-κB; and stimulates angiogenesis, proliferation, and controlled tissue remodeling. Its compact, electromechanical nature consolidates multiple therapeutic functions into a single platform, directly addressing the severe mass and volume constraints of exploration missions. Conclusion: CAPPJs represent a promising, multifunctional technology to address the compounded challenge of wound management in long-duration spaceflight missions. Targeted development and microgravity-specific research are recommended to advance its readiness for safeguarding crew health beyond Earth. Keywords: Cold Plasma Jets; Wound Healing; Space Medicine; Antimicrobial; RONS

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Letter of Explanation

Thank you for the opportunity to revise and resubmit my manuscript (formerly titled *An Overview of the Fabrication of a Cold Atmospheric Pressure Plasma Jet for Astronaut Health in Long Missions*) for consideration as a Review Article.

I appreciate the feedback provided by the editor and have addressed all comments in the revised version. Please let me know if revisions are satisfactory and appropriate. I am grateful for time and assistance and look forward to hearing from you.

Below is a list of revisions made in accordance with the comments/suggestions of the editor.

Comment:

It is submitted as a review article, but the materials and methods (which is a heading we do not use - Methods is preferred) seems to indicate that a device was constructed. That would imply a research article, but there are no results of any experiment reported.

Revision:

- The *Materials and Methods* section has been removed entirely from the manuscript. The focus is now exclusively on a review of existing literature. Replaced with *Technical Considerations: Device Configurations and Operational Parameters* which discusses general device rather than describing a specific prototype build.

Technical Considerations: Device Configurations and Operational Parameters

The efficacy of CAPPJ therapy is highly dependent on device configuration and operational parameters. Most research-grade and clinical devices utilize a dielectric barrier discharge (DBD) configuration, where a dielectric material separates the electrodes to prevent arc formation and ensure a stable, diffuse plasma at atmospheric pressure [11]. Common feed gases include helium, argon, or mixtures thereof, with gas flow rates and applied voltage (typically in the range of 1-40 kV at frequencies of 20-70 kHz) determining the plasma's chemical composition and intensity [12].

The choice of feed gas significantly influences the plasma plume characteristics and its biological effects. Helium, due to its low atomic mass and high thermal conductivity, tends to produce a diffuse, broad plume, which is useful for treating larger wound areas. Argon, with its higher atomic mass and lower ionization energy, creates a denser, more focused plasma plume, which may be beneficial for targeted disinfection or deeper tissue penetration. The reactive species generated, primarily RONS, are the key effectors of the biological response, and their concentrations can be tuned by adjusting the device parameters [16,17]. Figure 1 illustrates a constructed prototype laboratory-scale CAPPJ device, while Figures 2 and 3 depict design schematics and the visual differences in plasma plumes generated with different gas compositions.

Comment:

For a narrative review, any Methods that are discussed should relate only to the literature search parameters.

Revision:

- A new *Literature Search Strategy* section has been added, describing search terms, and inclusion criteria used to identify relevant studies.

Methods: A literature search was conducted. Search terms included "cold atmospheric plasma," "wound healing," "antimicrobial," "biofilms," "immunomodulation," and "space medicine." The review focuses on the bioactive components of CAPPJ, molecular to tissue-level mechanisms, and outcomes from clinical trials, contextualized within the constraints of spaceflight.

Comment:

I would suggest some effort be put into either more accurately presenting a narrative review on the effectiveness of this technology and its aerospace applications, OR redrafting the manuscript into a Technical Note (which would require max 3000 words and approx 10-12 references) that would describe just the technology.

Revision:

- The manuscript has been reorganized to adhere to a narrative review. The title has been revised to accurately reflect this refocused scope, and the content now only synthesizes published evidence on CAPPJ mechanisms, clinical outcomes, and potential spaceflight applications (device build excluded).

Cold Atmospheric Pressure Plasma Jets for Wound Management in Long-Duration Spaceflight A Review

Comment:

This is a complicated manuscript that needs some re-focus before being potentially sent out for peer review.

Revision:

- The structure has been streamlined. The *Technical Considerations* section now discusses general device parameters based on the literature, rather than describing a specific build (As mentioned as more in line w/ a Technical Note).
- Refocused abstract.

demonstrated terrestrial efficacy. **Objective:** This review synthesizes the mechanisms of Cold Atmospheric Pressure Plasma (CAPP) action and terrestrial clinical evidence to evaluate its potential as an integrated medical countermeasure for wound management in spaceflight. **Methods:** A literature search was conducted. Search terms included "cold atmospheric plasma," "wound healing," "antimicrobial," "biofilms," "immunomodulation," and "space medicine." The review focuses on the bioactive components of CAPPJ, molecular to tissue-level mechanisms, and outcomes from clinical trials, contextualized within the constraints of spaceflight.

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Background: Long-duration spaceflight, such as missions planned under NASA's Artemis program and to Mars, presents critical medical risks including impaired wound healing and infections from multidrug-resistant pathogens. The limitations of mass, volume, and medication shelf life make conventional wound care strategies inadequate. Cold Atmospheric Pressure Plasma Jets (CAPPJ) present a multifunctional, non-antibiotic therapeutic platform with demonstrated terrestrial efficacy. **Objective:** This review synthesizes the mechanisms of Cold Atmospheric Pressure Plasma (CAPP) action and terrestrial clinical evidence to evaluate its potential as an integrated medical countermeasure for wound management in spaceflight. **Methods:** A literature search was conducted. Search terms included "cold atmospheric plasma," "wound healing," "antimicrobial," "biofilms," "immunomodulation," and "space medicine." The review focuses on the bioactive components of CAPPJ, molecular to tissue-level mechanisms, and outcomes from clinical trials, contextualized within the constraints of spaceflight. **Results:** CAPPJ technology simultaneously eradicates pathogens, including biofilms, without inducing resistance; modulates dysregulated inflammation via pathways such as NRF2 and NF-κB; and stimulates angiogenesis, proliferation, and controlled tissue remodeling. Its compact, electromechanical nature consolidates multiple therapeutic functions into a single platform, directly addressing the severe mass and volume constraints of exploration missions. **Conclusion:** CAPPJs represent a promising, multifunctional technology to address the compounded challenge of wound management in long-duration spaceflight missions. Targeted development and microgravity-specific research are recommended to advance its readiness for safeguarding crew health beyond Earth.

Keywords: Cold Plasma Jets; Wound Healing; Space Medicine; Antimicrobial; RONS

Introduction: The Challenges of Space Medicine

Wound Healing in Space

Long-duration spaceflight induces pathophysiological changes that significantly impair the wound healing capabilities of astronauts. Microgravity disrupts cytoskeletal organization and cell migration [1,2], while radiation exposure compromises immune function and increases infection susceptibility [3]. The confined spacecraft environment promotes colonization by multidrug-resistant pathogens [4,5], creating an ecosystem that significantly increases the probability of wound complications [6,7]. With a round-trip duration predicted to be 1,000 days for proposed Mars expeditions [8], conventional wound management strategies become logistically impractical due to mass constraints, limited medical expertise, and medication shelf-life concerns [8,9,10].

Fundamentals of Cold Atmospheric Pressure Plasma Jets (CAPPJ)

Cold atmospheric plasma jets (CAPPJ) generate partially ionized gas at near-ambient temperatures through electrical discharges [11,12]. Nobel or inert gases such as helium or argon are typically used. These gases have a wide range of clinical applications and beneficial effects on biological processes [13,14,15]. The combination of electrical discharge and noble gas creates a unique bioactive environment containing reactive oxygen and nitrogen species (RONS: •OH, H_2O_2, •NO, O_2^-), electromagnetic fields, UV photons, and charged particles [16,17]. Like its terrestrial applications, CAPPJ's space utility stems from its multi-use functionality as it can simultaneously disinfect wounds, stimulate tissue regeneration, and manage inflammation [18,19,20].

Mechanisms of Action: Molecular to Tissue Levels

Antimicrobial Efficacy

CAPP exerts potent bactericidal effects through multiple synergistic mechanisms. Electroporation from electromagnetic fields and oxidation of lipid membranes by RONS causes membrane disruption [21]. DNA fragmentation by UV components and oxidative stress results in intracellular damage to microbes [22,23].

Biofilm penetration is enhanced by diffusion of RONS through glycocalyx matrices [24,25]. These simultaneous mechanisms resulting from cold atmospheric plasma could overcome antibiotic resistance, addressing a concern in spacecraft environments where drug-resistant pathogens could prevail [4,5,6].

Tissue Regeneration and Immunomodulation

CAPP stimulates wound healing through several biological mechanisms. CAPP modulates inflammation through reactive oxygen and nitrogen species (RONS)-dependent pathways [26,27]. CAPP-generated RONS activate the NRF2 antioxidant pathway [28,29,30], enhancing detoxifying enzymes like HMOX1 and NQO1 [31,32,33] to resolve chronic inflammation.

Simultaneously, CAPP exposure increases RONS levels, upregulating PPAR-γ and subsequently boosting IL-6 secretion [34], which is a key cytokine for immune cell recruitment. CAPP upregulates IL-6 and IL-8 in fibroblasts, promoting neutrophil and macrophage influx [35]. Additionally, CAPP-derived hydrogen peroxide (H_2O_2) regulates NF-κB, driving proinflammatory cytokine release and leukocyte attraction to initiate acute-phase healing [36,37,38]. Together, these mechanisms regulate immune activation while mitigating excessive inflammation in chronic wounds.

CAPP enhances angiogenesis and proliferation by upregulating key growth factors and activating critical signaling pathways. CAPP elevates HIF-1α, VEGF-A, and FGF-2 in keratinocytes [39,40],

while simultaneously activating eNOS [41], collectively improving microcirculation and endothelial function. CAPP also stimulates TGF-β1/2 [42], which, despite its pro-angiogenic properties, poses the risk of promoting scar formation [43] during remodeling [44,45,46]. Further, CAPP drives the HIPPO and YAP pathway [47,48], upregulating CTGF and Cyr61 (CCN1, a growth factor) [47,48] to mediate fibroblast-keratinocyte crosstalk, thereby facilitating extracellular matrix production and cellular migration essential for tissue regeneration.

CAPP optimizes tissue remodeling by enhancing extracellular matrix (ECM) reorganization while mitigating scarring [49]. CAPP directly upregulates Collagen Type I α (COL1a) synthesis in fibroblasts, strengthening nascent tissue [50]. Concurrently, it induces Matrix Metalloproteinase-1 (MMP1), which facilitates keratinocyte migration and ECM restructuring to accelerate re-epithelialization [51,52]. Crucially, CAPP also suppresses profibrotic factors such as TGF-β1, α-SMA, and COL1 in acute wounds, as demonstrated in rat models, resulting in significantly reduced scar width [53]. This dual promotion of structural repair while inhibiting excessive collagen deposition enables more functional and cosmetically favorable healing.

CAPP exerts potent antimicrobial effects. CAPP-generated RONS selectively disrupt bacterial membranes, causing significantly more damage to Gram-negative bacteria than Gram-positive species due to structural differences in their cell walls [54]. Additionally, CAPP upregulates endogenous antimicrobial peptides, with a 2-minute treatment boosting HBD-2 in human keratinocytes and repeated sessions elevating MBD-2/3 in murine skin [44]. Critically, CAPP eliminates pathogens without inducing resistance [55] by irreversibly damaging microbial DNA and proteins [56], offering a promising alternative to conventional antibiotics in chronic wound infections. Overall, optimization of voltage, gas, and exposure duration is critical for efficacy, as studies suggest dose-responsive outcomes.

Technical Considerations: Device Configurations and Operational Parameters

The efficacy of CAPPJ therapy is highly dependent on device configuration and operational parameters. Most research-grade and clinical devices utilize a dielectric barrier discharge (DBD) configuration, where a dielectric material separates the electrodes to prevent arc formation and ensure a stable, diffuse plasma at atmospheric pressure [11]. Common feed gases include helium, argon, or mixtures thereof, with gas flow rates and applied voltage (typically in the range of 1-40 kV at frequencies of 20-70 kHz) determining the plasma's chemical composition and intensity [12].

The choice of feed gas significantly influences the plasma plume characteristics and its biological effects. Helium, due to its low atomic mass and high thermal conductivity, tends to produce a diffuse, broad plume, which is useful for treating larger wound areas. Argon, with its higher atomic mass and lower ionization energy, creates a denser, more focused plasma plume, which may be beneficial for targeted disinfection or deeper tissue penetration. The reactive species generated, primarily RONS, are the key effectors of the biological response, and their concentrations can be tuned by adjusting the device parameters [16,17]. Figure 1 illustrates a constructed prototype laboratory-scale CAPPJ device, while Figures 2 and 3 depict design schematics and the visual differences in plasma plumes generated with different gas compositions.

Clinical Evidence: From Terrestrial to Spaceflight Applications

Terrestrial Efficacy

Clinical randomized controlled trials (RCTs) demonstrate that CAPP significantly enhances wound healing across diverse wound types. CAPP application achieves a 34–50.4% reduction in bacterial load in chronic wounds [57,58] and accelerates healing in diabetic foot ulcers, with studies

reporting 53–95.2% reductions in wound size [59,60]. For venous leg ulcers, twice-weekly CAPP yielded a 61.5% healing rate versus 25% in controls [59]. CAPP also improves secondary outcomes, including accelerated pain relief [57,61,62], with no serious adverse effects caused by CAPP device complications reported in any RCTs [59].

Spaceflight Applications and Research Needs

The translation of CAPPJ technology to spaceflight offers several distinct advantages. CAPPJ devices offer a single, compact platform capable of performing multiple critical functions: disinfection, inflammation modulation, and tissue regeneration stimulation. This significantly reduces the mass, volume, and complexity compared to carrying separate suites of antibiotics, antiseptics, advanced dressings, and growth factors required for comprehensive wound management over extended periods [8,9,10]. The elimination of drug shelf-life concerns is a major logistical advantage.

The potent, broad-spectrum antimicrobial action of CAPPJ, particularly its efficacy against biofilms and ability to kill without inducing resistance [24,25,55], is crucial for counteracting the heightened risk of infection from multi-drug resistant pathogens known to colonize spacecraft environments [4,5,6]. This offers a sustainable alternative where traditional antibiotic options are limited and resistance development is a severe threat.

CAPPJ's ability to stimulate key healing processes such as immune cell recruitment [35], angiogenesis [39,40,41], proliferation [47,50], and controlled ECM remodeling [51,52,53] demonstrates potential to overcome the wound healing impairments induced by microgravity and space radiation [1,2,3]. Its immunomodulatory effects [26,27,31], across mechanisms like the PPAR-γ/IL-6

pathway [34] and NF-κB regulation [36,37,38], could be vital for managing dysregulated inflammation in the space environment.

Future CAPPJ devices could be designed for relatively simple operation by crew medical officers or even astronauts themselves after appropriate training, reducing reliance on ground control-based expertise for wound management, a critical factor given communication delays during planned deep space missions. CAPPJ devices designed for space must take into consideration electrical safety, gas handling, and thermal management. Optimization to be operable in confined spacecraft settings with limited resources such as power and gas supply must be completed. Targeted development and microgravity-specific research are recommended to advance its readiness for safeguarding crew health beyond Earth.

Conclusion

Cold atmospheric pressure plasma jet technology represents a consolidated platform for addressing the documented challenge of impaired wound healing in long-duration spaceflight. This review has established that the mechanisms of CAPPJ action, which include eradicating pathogens without inducing resistance, modulating inflammatory pathways, and promoting essential tissue regeneration processes, align with the specific physiological deficits caused by the space environment.

Substantial evidence from terrestrial clinical trials supports the efficacy of this technology for improving healing in complex wounds. The compact and electromechanical nature of CAPPJ devices presents a distinct logistical advantage for exploration missions, as they perform multiple necessary therapeutic functions within a single system that meets stringent mass and volume constraints.

Integrating this technology into a spacecraft medical system requires a defined research pathway. Immediate priorities are the validation of its biological efficacy and operational safety in experimental models simulating microgravity and space radiation. Concurrent technical development must focus on ensuring device compatibility with spacecraft safety standards and resource limitations. Fulfilling these requirements is a necessary step in developing CAPPJ as a viable medical countermeasure for crew health maintenance on exploration missions beyond Earth orbit.

Acknowledgments

This work was conducted as part of a NASA (National Aeronautics and Space Administration) Office of STEM Engagement (OSTEM) internship at the Applied Chemistry Laboratory, Kennedy Space Center.

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Abbreviations

CAPPJ - Cold Atmospheric Pressure Plasma Jet

RONS - Reactive Oxygen and Nitrogen Species

OH – Hydroxide

H_2O_2 - Hydrogen peroxide

NO - Nitric oxide

O_2^- - Oxygen

UV – Ultraviolet

DNA - Deoxyribonucleic acid

NRF2 - Nuclear Factor Erythroid 2-Related Factor 2

HMOX1 - heme oxygenase-1

NQO1 - quinone oxidoreductase 1

PPAR-γ - Peroxisome proliferator-activated receptor gamma

IL-6 – Interleukin-6

IL-8 – Interleukin-8

NF-κB – nuclear factor kappa light chain enhancer of activated B cells

HIF-1α – Hypoxia-Inducible Factor 1-alpha

VEGF-A – Vascular Endothelial Growth Factor A

FGF-2 – Fibroblast growth factor 2

eNOS – Endothelial Nitric Oxide Synthase

TGF-β1/2 - Transforming Growth Factor beta 1 and 2

HIPPO – Salvador-Warts-Hippo pathway

YAP – Yes-Associated Protein

CTGF – Connective Tissue Growth Factor

Cyr61 – Cysteine-rich angiogenic inducer 61

CCN1 - Cellular Communication Network 1

ECM - Extracellular matrix

COL1a - Collagen Type I α

MMP1 - Matrix Metalloproteinase-1

TGF-β1 – Transforming Growth Factor beta 1

α-SMA – Alpha-smooth muscle actin

COL1 – Collagen Type I α

HBD-2 – Human beta-defensin 2

MBD-2/3 – Methyl-CpG Binding Domain Protein 2 and 3

RCTs - Randomized controlled trials

HV+ - high voltage

Figure 1 Image of the assembled Cold Atmospheric Pressure Plasma Jet (CAPPJ) on a clamp stand, labeled with its components.

Figure 2 Design overview of the assembled Cold Atmospheric Pressure Plasma Jet (CAPPJ).

Figure 3 (A) Pure Helium gas flow. (B) Helium and Argon mixture gas flow. (C) Pure Argon Gas Flow.

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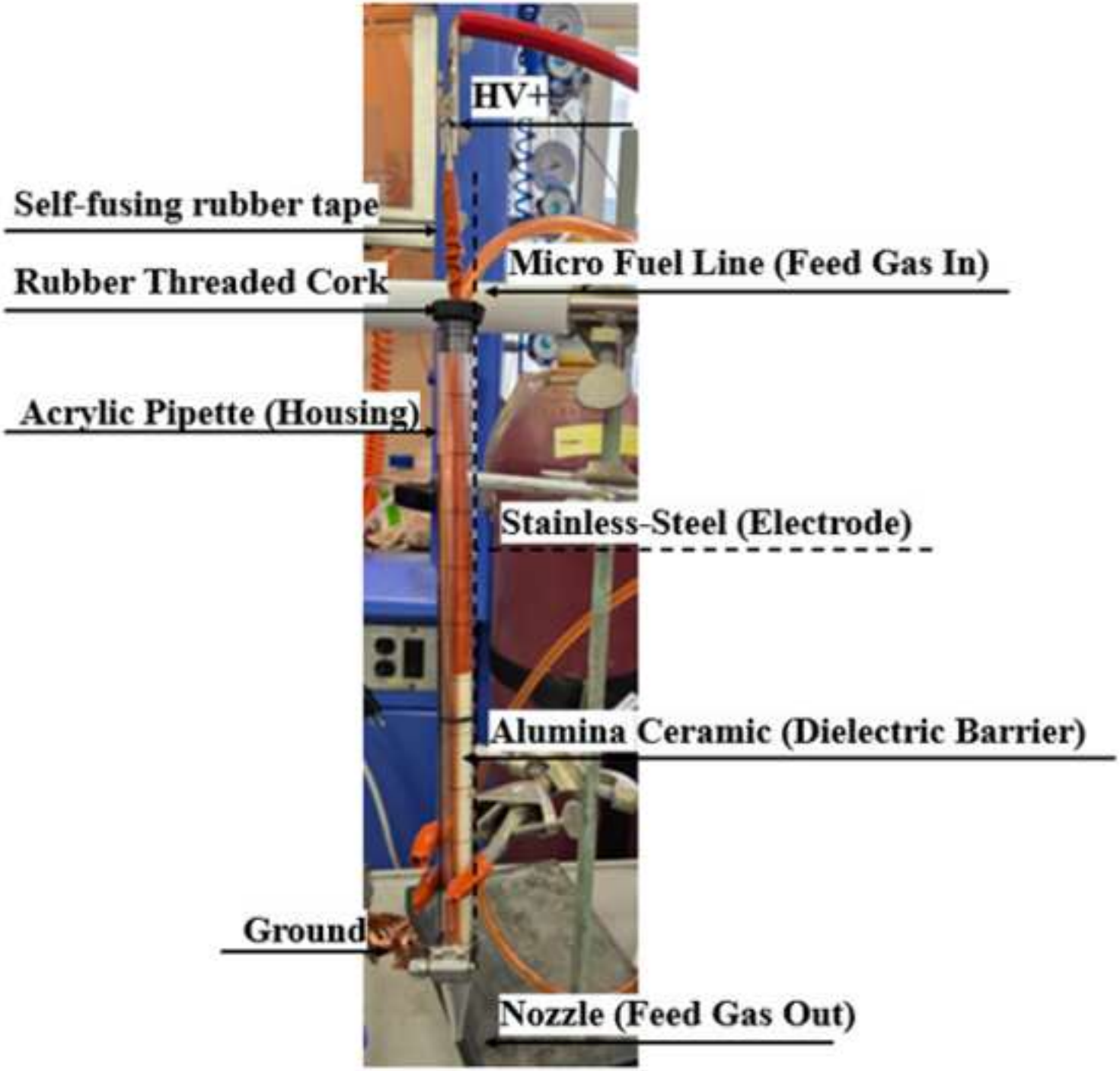

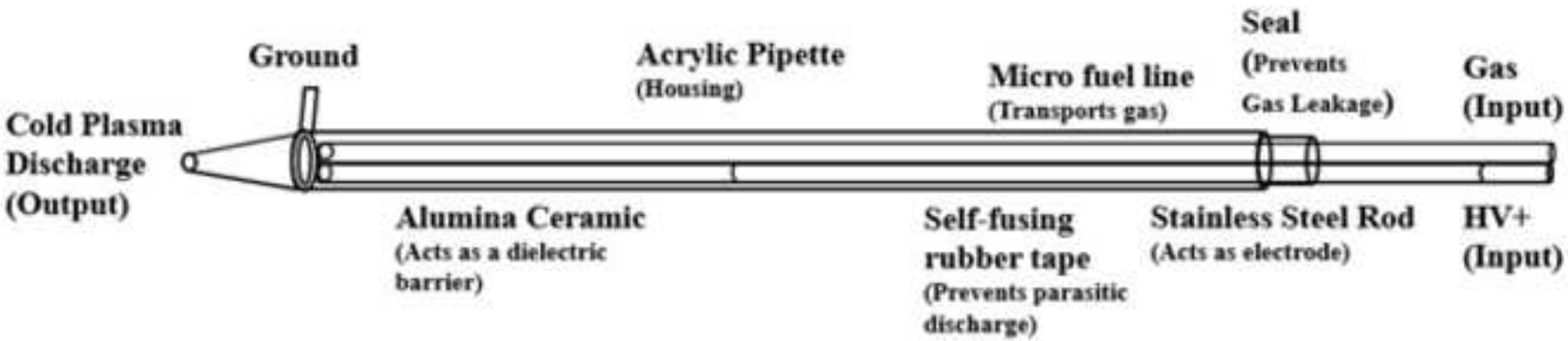

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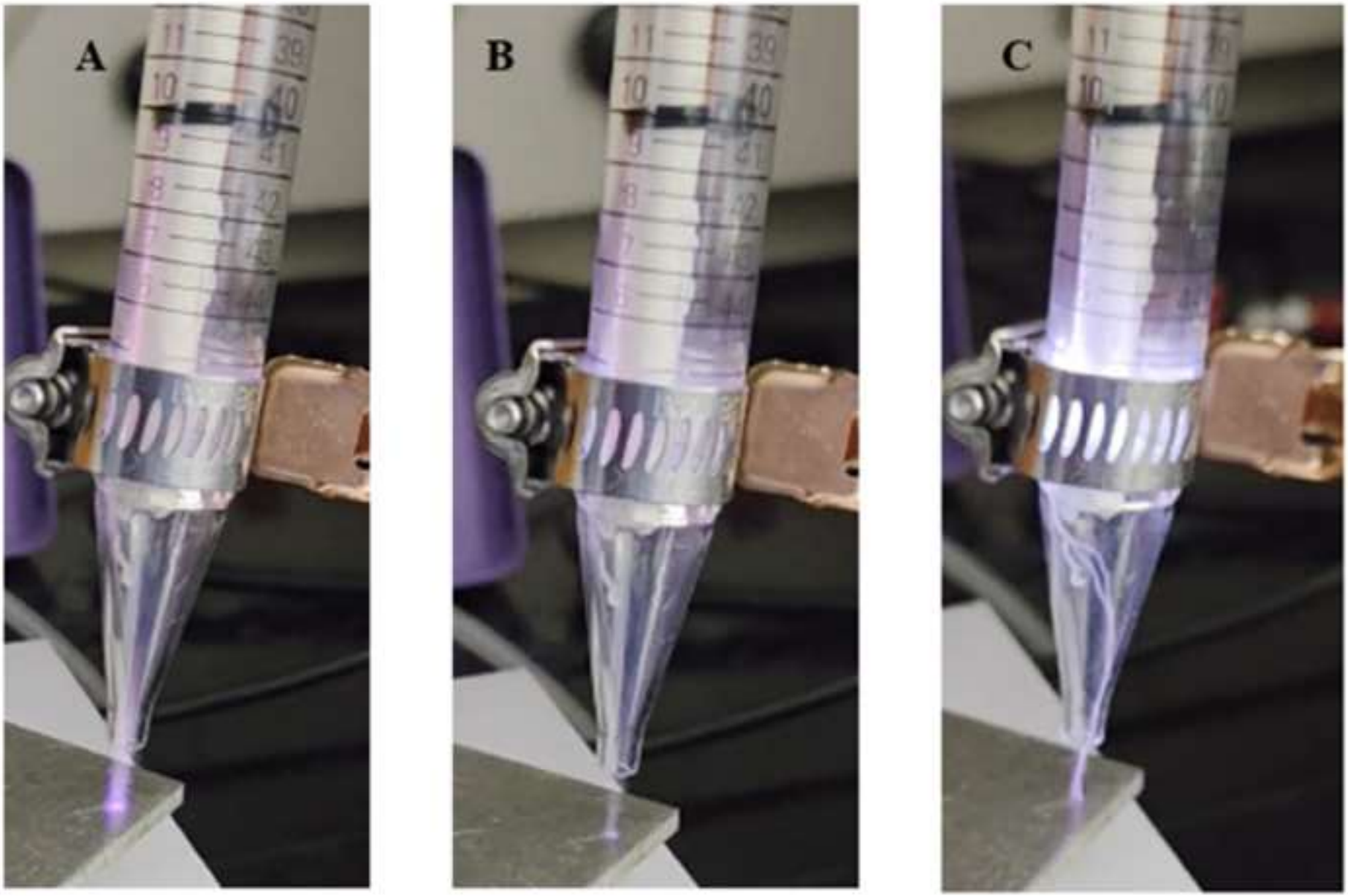
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